

Game-Theoretic Industrial Control Systems Defense using Cyber Kill Chain Model and MITRE ATT&CK

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ABSTRACT

This research brings up the vulnerability of Static Defenses in Industrial Control Systems (ICS) against rational, learning adversaries and shifts the focus from simple maintenance costs to Industry 5.0 Resilience metrics. By using Game Theory (Mixed-Strategy Nash Equilibrium) the Cyber Kill Chain and MITRE ATT&CK techniques in a Python-based simulation engine we evaluate three defensive taxonomies: Static, Hardened, and Adaptive (Dynamic).

Baseline simulations confirms that Static Defense is ineffective, yielding a 98% attacker success rate against a Bayesian learning opponent. Comparative analysis reveals that while a Hardened Defense reduces breaches it generates excessive operational noise (False Positives > 0.50) and extends the Mean Time to Detect (MTTD) by forcing attackers into stealth. The Adaptive Defense strategy proved to be the best choice of the three strategies. It established a strategic "Choke Point" at the Exploitation stage and successfully breaks the kill chain while having lower defense costs and operator alert fatigue. The findings empirically demonstrate that dynamic, game-theoretic strategies can give the superior balance of Sustainability, Human-Centricity and Resilience required for modern industrial security [5].

Keywords

Cyber Kill Chain (CKC); MITRE ATT&CK; Industrial Control Systems (ICS); Predictive Maintenance (PdM); Cybersecurity Simulation; Game Theory; Monte-Carlo Modelling; Intrusion Detection; Resilience

1.INTRODUCTION

Predictive Maintenance (PdM) and eMaintenance systems improve efficiency in Industrial Control Systems (ICS) but they

also create new cyber vulnerabilities [1]. By relying on continuous sensor data, network connectivity, and automated decision-making, these systems expand the attack surface and make failures harder to detect. While this project initially aimed to find the financial cost of cyber attacks on ICS assets the literature shows that such impacts are better captured through resilience rather than single loss values [2]. In industrial environments, security is not a question of whether a system is "safe" or "hacked" but how well it maintains and restores functionality during and after an attack. This impact can be measured as performance loss over time using the resilience curve [2].

A major weakness in current ICS security is the use of static defense policies. These policies assume that attacks are predictable and that fixed rules are sufficient to stop them. In PdM systems this assumption often fails. For example, attackers can manipulate sensor data so that malicious activity appears as normal equipment wear, delayed maintenance or faulty measurements. Such attacks can bypass rule-based detection while still degrading system performance. At the same time can strict security controls increase operator workload and false alarms, leading to alert fatigue and unsafe workarounds in human-centered Industry 5.0 environments [2], [3].

This creates a clear research gap. Many defensive approaches focus on hardening systems against known threats but they do not address adaptive attackers who learn and change strategies over time. To explore this gap, this project models ICS cybersecurity as a strategic interaction between attacker and defender rather than as a fixed rule set. Using a simultaneous-move game and Mixed-Strategy Nash Equilibrium (MSNE), the study examines whether adaptive and probabilistic defense strategies can improve resilience while remaining operationally sustainable [3].

2.BACKGROUND / RELATED WORK

OT and ICS differ from traditional IT systems [5] because they directly control physical processes with strict safety and availability requirements. In PdM environments the attacks can remain hidden by blending into normal maintenance data, making detection especially difficult. Therefore security failures may appear as technical faults rather than cyber incidents.

To analyze these threats this project uses the Cyber Kill Chain (CKC) as a structural framework. Each stage of the CKC is treated as a decision point where attackers and defenders choose actions under uncertainty. These stages are populated with real techniques from the MITRE ATT&CK for ICS knowledge base,

such as modifying controller logic or disabling safety functions to reflect realistic industrial attack behaviour.

Some prior work models cybersecurity interactions using leader-follower (Stackelberg) formulations where the defenders strategy is fixed before the attacker responds. However, real ICS attacks occur under uncertainty, where neither side knows the others next move. To better represent this setting, this project uses a simultaneous-move model and applies Mixed-Strategy Nash Equilibrium to generate defensive strategies that include uncertainty. These strategies aim to reduce the predictability for the attacking-side, slow attacker learning and limit overall performance loss.

By linking these strategic outcomes to the resilience curve, the project evaluates succeeded attacks and how much damage they cause over time and how quickly the system recovers. This connects abstract game-theoretic reasoning to practical resilience assessment in industrial systems.

3.METHODOLOGY

This project creates a computational simulation to quantify the resilience of Industrial Control Systems (ICS) against rational adversaries. Unlike traditional vulnerability assessments that view security as a binary state (safe versus compromised), this methodology models the attack-defense dynamic as a continuous, strategic interaction. The simulation pipeline transitions from abstract signal modelling to a Game-Theoretic State simulation, uses the Cyber Kill Chain (CKC) to structure the conflict and Mixed-Strategy Nash Equilibrium (MSNE) to solve for optimal decision-making under uncertainty.

3.1. Game-Theoretic Formulation And The Cyber Kill Chain

The Cyber Kill Chain is modeled as a set of strategic decision points rather than a linear process. Each stage is treated as a non-cooperative, simultaneous-move interaction to capture the uncertainty of cyber operations where neither side can predict the others actions in advance.

To populate these abstract stages with realistic industrial constraints the model uses the MITRE ATT&CK for ICS knowledge base. Specific attack techniques, such as "Modify Controller Tasking" (T0821) or "Change Operating Mode" (T0858) are parameterized with base success probabilities ($P_{success}$) and detectability coefficients (λ). Actions in the early stages like Reconnaissance carry low potential rewards, while later stages such as Actions on Objectives are assigned a stage multiplier (μ_k) that increases utility (up to 1.0), reflecting the higher value of executing physical damage.

The interactions are structured as 3×3 payoff matrices, pitting three Attacker strategies; Stealthy (low risk, low reward), Standard (balanced), and Aggressive (high risk, high reward) against varying Defender monitoring levels. The simulation solves these matrices using the Expected Utility Property where a rational attacker does not act randomly but maximizes their benefit based on the probability of outcomes. The result is made using the nashpylibrary to find the Mixed-Strategy Nash Equilibrium (MSNE) which is the state where neither player can increase their payoff by changing their strategy.

3.2. Simulation Engine And Agent Logic

The simulation engine is built in Python and controls the behaviour of two distinct agents: a Bayesian Attacker and a Defender(configured as either Static or Dynamic).

The Bayesian Attacker: To simulate a learning adversary, the attacker maintains a belief state θ_t representing the probability that the defender is "High Capability" (Strong). Initially set to a neutral belief ($\theta_0 = 0.5$), this variable is updated using Bayesian Learning after every interaction. If an attack is blocked, the attacker updates their belief using Bayes rule (θ_{t+1}) which shifts their assessment toward a "Strong Defender" and mathematically incentivizes them to switch to Stealthy strategies in next steps. Evasions lower θ_t and encourages Aggressive tactics.

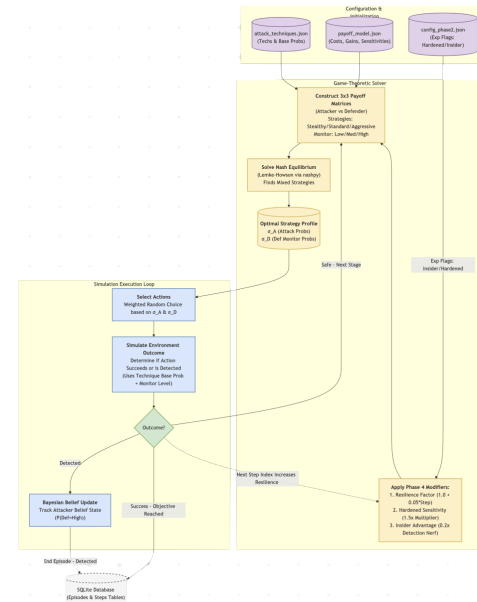


Figure 1. Architecture of the Game-Theoretic State Simulation

Defender Taxonomies:

- **Static Defense:** This baseline model uses fixed policies (e.g., a constant probability of blocking) regardless of the attackers actions. This serves as the control group to demonstrate the vulnerability of rigid security postures against learning adversaries.
- **Dynamic (Adaptive) Defense:** This strategy implements a reactive history buffer. If an intrusion is detected at Stage k , the defender "learns" and temporarily boosts the blocking probability and monitoring granularity at Stage $k+1$.

Insider Threat Scenario: To account for Human-Centric risks in Industry 5.0 a specialized scenario was constructed. In this mode the detectability coefficient is reduced by approximately 80%, simulating the trust and access granted to trusted personnel.

3.3. Data Architecture And Evaluation Metrics

The project uses a hierarchical (one-to-many) SQLite database schema instead of file logging. Experiments store configuration settings, Episodes capture individual runs, and Steps record detailed strategy and belief updates. The system is evaluated using resilience rather than success or failure and measured as the area under the curve over time.

Evaluation is based on three indicators:

Total Cost of Defense (TCD) that aggregates defense costs, downtime, and false-positive penalties; Attacker Survival Rate, which measures how far an attacker progresses through the Kill

Chain and captures the Domino effect of cascading failures across stages; and Signal-to-Noise Ratio which shows how blocking rates and false positives affect operator workload.

4. Strategic Game-Theoretic Framework

This section formalizes the interaction between the Industrial Control System (ICS) defender and the adversary using a computational game-theoretic model. Unlike static vulnerability assessments, the framework treats the Cyber Kill Chain (CKC) as a series of interdependent, simultaneous-move games. At each stage, optimal strategies are computed in real time using a Mixed-Strategy Nash Equilibrium (MSNE) [1], [10].

4.1. Game Structure

The simulation models cyber operations as a finite, state-based game played across the seven stages of the CKC. At each stage $k \in 1...7$ from Reconnaissance to Actions on Objectives the attacker and defender must simultaneously select an action without knowledge of the opponents current choice. This "fog of war" is modeled as an Imperfect Information Game.

The interaction is controlled by the specific parameters defined in the system configuration (payoff_model.json) where the objective value (V_k) increases non-linearly as the attacker progresses closer to the physical process ranging from 1,000 Monetary Units (MU) at Reconnaissance to 15,000 MU at Actions on Objectives [10].

4.2. Players And Agent Taxonomies

The simulation defines two primary agents each with distinct rationality and capabilities.

The Bayesian Attacker (A): The rational attacker learns while seeking to max the utility of reaching the final CKC stage. The attacker maintains a belief state θ_t which means that the defender has "High Capability." This belief starts at $\theta_0 = 0.5$ (neutral) and updates according to Bayesian direction based on the outcome of each stage [3].

Insider Threat Variant: The trusted insider experiments has a mathematically reduced detectability coefficient ($\lambda_{insider} \approx 0.2 \lambda_{standard}$) [2].

The Defender (D): The defender wants to minimize the Total Cost of Defense (TCD) which is the sum of operational costs, false positive penalties (C_{FP}) and the potential impacts of a successful breach (C_{impact}). The simulation implements three distinct defender taxonomies [1]:

- **Static Defense:** Is defined by simple fixed policy vector.
- **Hardened Defense:** High detection ($\lambda = \text{high}$) but also high operational cost ($C_{active} = 80$ MU) and increased false positive FP rates.
- **Dynamic (Adaptive) Defense:** Uses history to change detection rates and are boosted in stage $k+1$ if strange activity is detected in stage k .

4.3. MITRE ATT&CK Mapping

To connect the game to ICS trade-offs the action simulation is mapped to specific MITRE ATT&CK for ICS techniques.

Attacker Action Set (As): At every stage the attacker chooses from three intensities:

- **Stealthy (a1):** Low-risk actions (e.g., Network Sniffing - T0842). Yield lower immediate utility (40 MU) but has lower probability of detection.
- **Standard (a2):** Balanced actions (e.g., Modify Controller Tasking - T0821). Yields moderate utility (100 MU).

- **Aggressive (a3):** High-impact actions (e.g., Change Operating Mode - T0858). Yields maximum utility (500 MU) but carries a high detection penalty (-150 MU).

Defender Action Set (Ds): The defender chooses between:

- **Passive Monitoring (d1):** Low-cost ($C_{passive} = 5$ MU) with small detection probability.
- **Standard Monitoring (d2):** Balanced ($C_{standard} = 25$ MU). Increases detection probability without the high disruption risk of active blocking.
- **Active Defense (d3):** High-cost defense ($C_{active} = 80$ MU) involving blocking or segmenting. Highest chance of stopping attacks but carries operational disruption risk.

4.4. Payoff Matrices And Utility Functions

The core logic of the simulation is the dynamic construction of a 3×3 payoff matrix at each simulation step. The utility functions are derived from the payoff_model.json configuration [3].

Attacker Utility: The expected utility for an attacker choosing action a_i is defined as:

$$U_A(a_i) = P_{succ}(a_i) \cdot V_k - P_{det}(a_i) \cdot C_{penalty}$$

Where V_k is the objective value of the current CKC stage and $C_{penalty}$ is the cost of detection (e.g., loss of infrastructure access).

Defender Utility: The defenders utility is an inverse cost function:

$$U_D(d_j) = - [C_{def}(d_j) + P_{fail}(d_j) \cdot C_{impact} + P_{FP}(d_j) \cdot C_{maint}]$$

Where C_{maint} represents the cost of long downtimes and investigating false alarms (set to 500 MU) as the "human-centric" cost of operator fatigue [2].

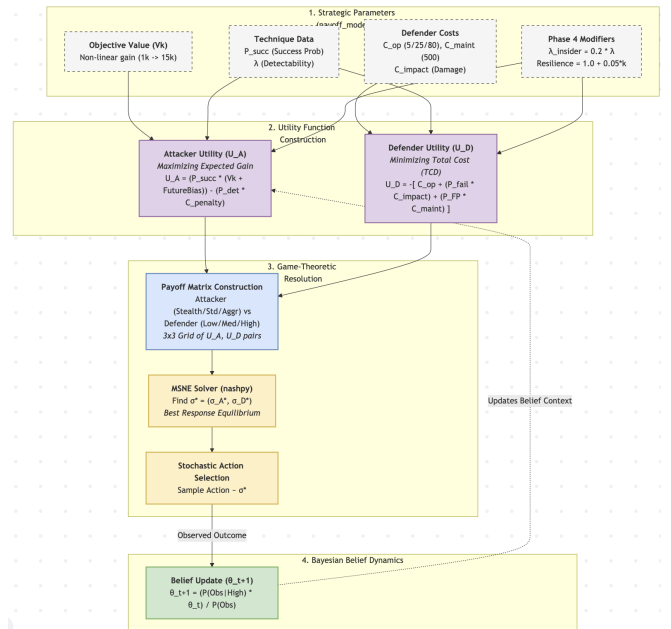


Figure 2. Mathematical Engine.

4.5. Equilibrium Solver (MSNE)

Unlike earlier qualitative models this simulation solves the game at every step. The engine uses the nashpylibrary to calculate the Mixed-Strategy Nash Equilibrium (MSNE). The solver finds the probability distribution over actions $\sigma^* = (\sigma A^*, \sigma D^*)$ such that neither player can improve their payoff by unilaterally deviating.

$$\sigma^* = \text{NashSolver}(\text{Matrixpayoff})$$

This calculated equilibrium σ^* determines the probabilistic behaviour of the agents for that specific step. For example, if the MSNE returns $\sigma A = [0.8, 0.2, 0.0]$, the attacker will choose Stealthy 80% of the time and Standard 20% of the time. These simulation results follow the "best response" dynamics of a rational conflict [1].

4.6. Bayesian Belief

To capture the sequential nature of learning, only the Attacker updates their belief θ_t using Bayes Rule after observing the Defenders action (Block or Pass).

$$\theta_{t+1} = \frac{P(\text{Obs} | \text{HighCap}) \cdot \theta_t}{P(\text{Obs})}$$

If an attack is blocked, the likelihood ratio shifts θ toward 1.0 (Strong Defender). As θ increases, the Nash Solver automatically adjusts the attackers mixed strategy toward Stealthy moves, because the expected utility of Aggressive actions decreases against a high-capability opponent.

4.7. Dwell Time

The simulation also takes into effect an increasing resilience depth. As the attacker progresses through the Kill Chain stages 0-7 a Resilience Factor is applied (increasing detection sensitivity by 5% per step) to simulate that dwell time might affect an adversaries increasing chance of being detected since anomaly detection systems and chances or the time it takes for operators to investigate a system fault code is improved over time.

5. RESULTS

The simulation executed 200 Monte-Carlo episodes for each of the four experimental scenarios: Static Defense (Baseline), Hardened Defense, Insider Threat, and Dynamic (Adaptive) Defense. The results were aggregated via the SQLite database and analyzed using the "Area Under the Curve" resilience metric and Total Cost of Defense (TCD).

5.1. Baseline Failure Of Static Defense

The initial control scenario evaluated a Static Defender against a rational Bayesian Attacker. Consistent with the theoretical prediction of the "Expected Utility Property," the static strategy resulted in a defense failure with a 98% attacker success rate.

The simulation data has a clear behavioural evolution that in the early Reconnaissance and Weaponization stages, the attacker dominantly selected Stealthy strategies (a1) to minimize detection risk while building a belief state about the defender. As the attacker successfully evaded detection, their belief in a "Strong Defender" (θ_t) dropped from 0.5 to approximately 0.35. This confidence emboldened the adversary to switch to Aggressive tactics (a3) in the final Actions on Objectives stage to maximize their utility. This confirms that a static posture, regardless of its monitoring level is highly vulnerable to a learning adversary who can identify and exploit fixed patterns.

5.2. Resilience Choke Point

In the comparative phase, the **Attacker Survival Rate** (Figure 3) demonstrates the structural superiority of the Dynamic Defense.

While the Hardened Defense (square line) successfully lowered the total success rate compared to the baseline, it still allowed the attacker to survive deep into the Kill Chain, maintaining an >80% survival rate through the mid-game stages. In contrast, the Dynamic Defense (circle line) created a distinct strategic "Choke Point" at the Exploitation stage, sharply reducing attacker survival to ~55%.

This steep drop indicates that the Dynamic Defense successfully disrupted the attackers Bayesian learning. By reacting to early indicators in the Delivery phase with increased blocking probability in the Exploitation phase, the adaptive strategy forced the attacker into a high-risk decision space where their expected utility calculations failed and led to an early stop of the kill chain.

5.3. Cost-Resilience Trade-Off

The economic analysis (Figure 4) validates the Industry 5.0 sustainability hypothesis. The Cost-Resilience Trade-Off plot visualizes the relationship between the Total Cost of Defense (TCD) and the Impact (Attacker Utility).

The Dynamic Defense occupies the "Optimal Zone" (bottom-left quadrant), achieving the lowest Attacker Utility (~10,000 MU) with a minimized Defense Cost (~3,100 MU). Conversely, the Hardened Defense incurred a significantly higher cost (~3,600

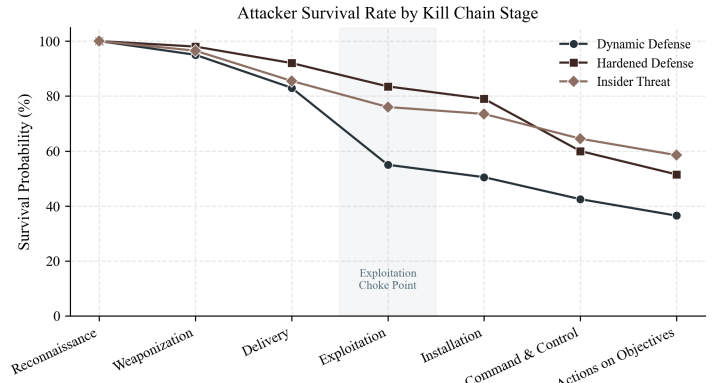


Figure 3. Attacker Survival (by CKC stage).

MU) without a proportional reduction in impact. This empirical evidence proves that "more security is better" is wrong and shows instead that "smart" adaptive security yields a better return on Investment (ROI) and lower resource consumption.

5.4. Operational Load And Signal-To-Noise

To address the Human-Centric requirements of Industry 5.0, the simulation measured the Signal-to-Noise Ratio (Figure 5) where Blocking Rates are against False Positives.

The Hardened Defense generated a False Positive Rate > 0.50, placing it squarely in the "Alert Fatigue Zone". In a real-world ICS environment such a high volume of false alarms would likely lead operators to ignore warnings or bypass security controls, a phenomenon identified as the "weakest link" in related literature. The Dynamic Defense, however, maintained a high Blocking Rate (>84%) while keeping the False Positive Rate significantly lower (<0.35). This confirms that adaptive strategies cry wolf less and follows with the "Human-Centric" design philosophy.

5.5. The Insider Threat And Mean Time To Detect (MTTD)

The Insider Threat scenario (starting at Exploitation with high trust) highlighted the limits of technical controls. Despite the advantage, the Dynamic Defense still outperformed the Static baseline, though with reduced efficacy compared to external threats.

An important yet counter-intuitive finding emerged regarding the Mean Time to Detect (MTTD) (Figure 6). The Hardened Defense showed a slower MTTD than the Dynamic Defense. This apparent paradox is however explained by the Game-Theoretic solver. When facing a "Hardened" wall, the Rational Attacker opted for extreme stealth and chose to dwell in the system for longer periods to avoid detection. The Dynamic Defense however, by creating uncertainty forced the attacker into making suboptimal and aggressive errors that lead to a faster detection.

6. DISCUSSION

These findings redefine the role of the Cyber Kill Chain (CKC) in Industry 5.0 and confirms that resilience is not a product of static "hardening" but of strategic adaptation. [2].

Attackers enter the same attack landscape as the defender, only from the opposite side. Just as attackers exploit the environment to their advantage, defenders can use the landscape strategically and by shaping attacker behavior over time, they may force adversaries to favor lower-risk strategies.

6.1. Attacker Success And Monitoring

The baseline simulations exposed the fundamental fragility of static defense policies. The near total collapse of the Static Defense is not a configuration failure but a mathematical outcome of Bayesian decision theory [3]. As the attacker observes a fixed policy their belief about defender capability (θ_t) drops and validates the environment as "safe" and allowing predictable gaps to be exploited. This demonstrates that security through obscurity fails against a learning opponent [1].

In sharp contrast the Dynamic Defense adds uncertainty for the attacker. By probabilistically shifting its defense posture in response to early warning signals, the model degraded the attackers ability and reduced expected gains. The Cost-Resilience Trade-Off (Figure 4) places the Dynamic Defense in the "Optimal Zone" achieving the lowest attacker utility (~10,000 MU) at the lowest cost (~3,100 MU). For an ICS analyst this offers an interesting insight: the uncertainty introduced by adaptation functions as a "virtual cost" for the attacker. It forces them to abandon high-yield aggressive strategies even when the defender is not actively blocking deterring the attack through mathematical risk rather than brute force [3].

6.2. CKC Stage Vulnerabilities

Traditional security models often treat the Cyber Kill Chain as a linear pipe where risks accumulate equally. This project shows it is better seen as a funnel where some stages matter more than others. The Attacker Survival Rate (Figure 3) identifies the Exploitation stage as the decisive "Bottle Neck". While the Hardened Defense allowed over 80% survival through mid-game, the Dynamic Defense reduced it sharply to ~55% at this stage. This indicates that hardening early stages, like Reconnaissance, has limited impact. The "Area Under the Curve" analysis shows that resilience is maximized by stopping attacks before physical processes are affected which aligns with Industry 5.0 definition of resilience as the capacity to withstand and recover [2].

6.3. Effectiveness Of Game Model

Perhaps the most significant finding for OT strategy is the Mean Time to Detect (MTTD) paradox shown in Figure 6. The Hardened Defense despite being the most expensive and theoretically "secure" resulted in a slower time to detect a True Positive than the Dynamic Defense. To clarify, in a real ICS environment a high FP rate often means that the "Time to Detect" is technically instant (alerts are firing everywhere) but the "Time to Truth" is infinite because they are ignored.

This finding challenges the core tenet of traditional "Defense in Depth." A rational Bayesian Attacker, facing a high-probability blocking wall (High Monitoring) pivots to Stealthy strategies (a1) to preserve utility. This extends their "dwell time," allowing them to persist undetected within the ICS environment, potentially establishing persistence (T0889) without triggering alarms [3]. By contrast the Dynamic Defense creates a strategic "baiting" effect. Its lower initial resistance encourages the attacker to attempt Standard or Aggressive moves which carry higher detection probabilities. In game-theoretic terms, the Dynamic Defense forces the attacker to commit unforced errors, turning their own utility max into a vulnerability [1], [3].

6.4. MITRE ATT&CK

The integration of specific MITRE ATT&CK techniques connected the mathematical model in ICS system reality, especially about the "Human-Centric" part of Industry 5.0. The Signal-to-Noise analysis (Figure 5) highlights a critical disconnect between theoretical security and human capability [4].

The Hardened Defense generated a False Positive Rate > 0.50, placing it in the "Alert Fatigue Zone." In a real-world control room, this is a failure state; a defense that generates more noise than signal will inevitably be bypassed or ignored by human operators [2]. This directly addresses the "Lack of Skill" and workforce shortages identified in current literature [5], as reducing the cognitive load on operators is essential when specialized personnel are scarce.

The Dynamic Defense with a False Positive Rate < 0.35 validates that an optimal strategy must respect cognitive limits. By using the Nash Equilibrium parameters with operators behaviour towards alerts we saw that the adaptive model sees to that alerts remain helpful. This confirms that Game Theory can be used to not just optimize for victory but also for operational sustainability [1], [2].

6.5. Implications And Recommendations

The study provides several key insights for ICS cybersecurity. Monitoring policies must have a balance between detection rates and operational cost where medium-level monitoring often provides the most practical trade-off. Stage-specific defenses are critical, early attacks are hard to detect while later interventions reduce attacker success. The simulation shows how using MSNE to choose strategies together with organized data analysis can help design and prioritize defenses.

6.6. Limitations

Despite the insights gained the study has limitations that should be acknowledged. The probability and payoff values used in the simulation are simplified and hypothetical and real-world ICS attacks may involve more complex or adaptive strategies. The model assumes rational behaviour from both attacker and defender which may not always reflect a real scenario. Only three monitoring levels were tested and exploring a broader range of policies could refine conclusions. The simulation database use

synthetic data and using real industrial log data would bring accuracy, validation and reality to the models results.

7.CONCLUSION AND FUTURE WORK

This project successfully combined Game Theory, the Cyber Kill Chain (CKC) and the MITRE ATT&CK framework to build a working simulation of an attacker-defender game in ICS security. By moving from a simple checklist to a dynamic game the project provides a clear way to test Industry 5.0 strategies.

The results from the 200 simulation runs showed some clear and important patterns. One of them being the Static Defense that failed completely. The rational attacker succeeded 98% of the time because they could easily learn from the fixed defense pattern and bypass it.

I also found that simply "Hardening" the system (Active Defense) does not work as well as expected. While it stopped more attacks, it created huge problems with noise. The False Positive Rate was over 50%, which in the real world means operators would just stop listening to most alarms. Even stranger, the Hardened Defense actually had a slower detection time (MTTD) than the Dynamic Defense. This is because the strong defense forced the attacker to be more stealthy and hide for longer but also still survived longer.

The winner from the different strategies was the Dynamic (Adaptive) Defense. By changing its strategy based on history, it created a "Choke Point" at the Exploitation stage. It broke the attack chain in the middle while keeping costs low. This validates the core hypothesis that resilience is not achieved by building higher walls (Robustness) but by adapting to the attack landscape.

7.1.Future Work

My current model is somewhat successful but there are many ways to make it more useful in a real world ICS scenario.

The current model assumes the attacker is "perfectly rational" (using pure math). Future work should use Deep Reinforcement Learning (DRL) to be able to follow resilience levels continually and see what strategies works best to withstand, bounce back and improve over time. DRL is a type of Machine Learning that learns by trial and error more like a human and would allow for a simulation where there are attackers who make mistakes or act irrationally under stress which is more realistic than a perfect mathematical solver.

I also want to move away from abstract "Monetary Units" and connect the simulation to a Digital Twin. Instead of just calculating a probability, the game could run on a real test network topology (like Mininet). This would let us see if the "Choke Point" effect still happens when we deal with real network traffic and delays.

The Insider Threat scenario could be improved. Right now, the Insider just gets a bonus to their stealth. In the future I would like to turn this into a "Signaling Game" where the defender could set traps (like Honeypots) to trick the insider. The defender would actively try to confuse the attacker making them believe the system is safe when it is actually a trap.

7.2.Summary

This project proves that in the age of Industry 5.0 the traditional concept of "Cyber Security" has become old and vulnerable. We can no longer build walls high enough to keep a rational and learning attacker out. Instead, we must shift our entire philosophy from Prevention (stopping the attack) to Resilience (surviving the attack).

The simulation results reveal a big strategic truth that uncertainty is a weapon. The Static Defense failed not because it was weak but because it was predictable. The Hardened Defense failed because it tried to be strong everywhere and created so much noise that it blinded the human operators. The Dynamic Defense succeeded because it introduced risk into the attackers calculation. By changing the rules of the game in real-time it forced the adversary to second-guess their own strategy which led to a psychological "Choke Point" that broke the attack chain.

This framework offers more than just a simulation; it offers a new way of thinking. It demonstrates that we do not need perfect security but we just need to be unpredictable enough to make the cost of attacking us higher than the reward.

So, in this projects "attack" to find the truth about the true costs of static defenses, the harsh reality is that if you are still relying solely on historical vulnerability patching and compliance checklists, you are not playing chess anymore but simply waiting for the checkmate and your opponent is already three moves ahead.

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10. Simulation Configuration Files (payoff_model.json, config_phase2.json), Project Files.
11. Project Simulation Database (phase2.db) and Output Figures 1–4.

REMAINING FIGURES

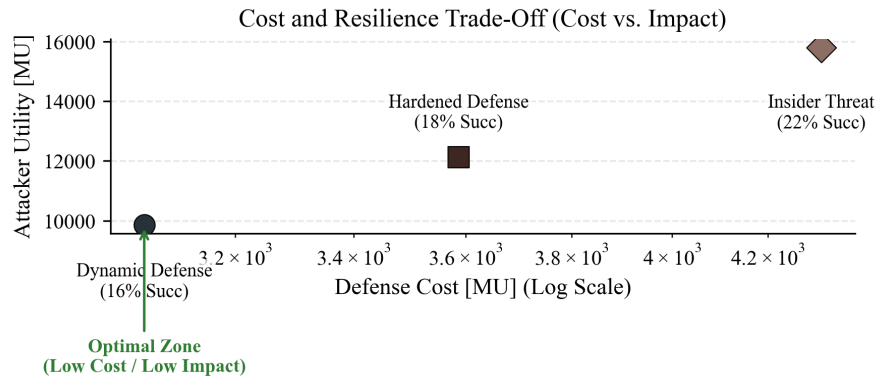


Figure 4. Cost Resilience Trade-Off (Cost vs. Impact).

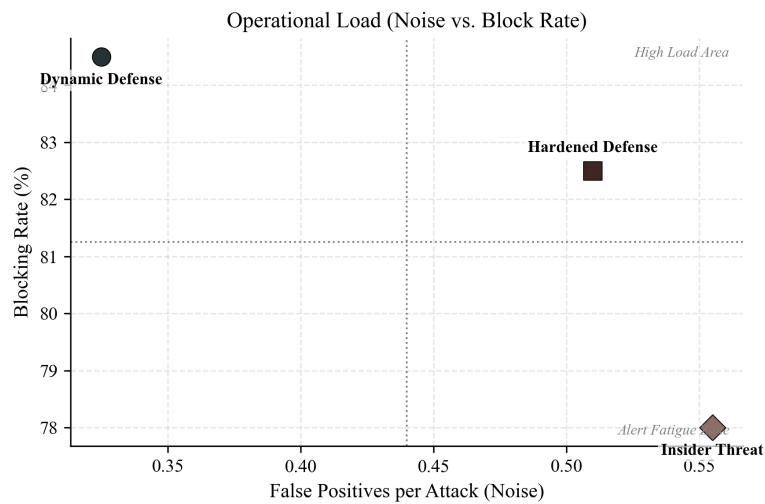


Figure 5. Operational Load (Noise vs. Block Rate).

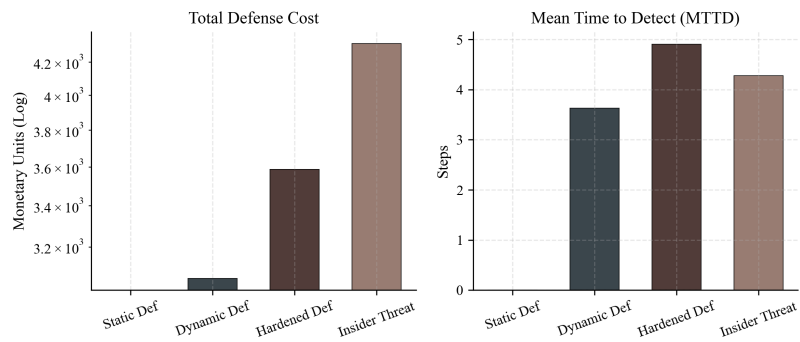


Figure 6. Total Defense Cost and Mean Time To Detect (MTTD).